BrandLens

Pre-publication brand governance: review content against approved guidelines, facts and assets before it goes live.

Problem statement by Inkloom AI | We Code Coders Hackathon | Round 1 - The Plan

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In one line: BrandLens checks every draft against the organisation's own approved source material, explains each issue with the exact guideline it comes from, suggests a fix that keeps the creator's intent, and leaves every approval to a human. It also shows managers which mistakes keep repeating and which guidelines are too vague.

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1. Understanding of the Problem Statement

Organisations no longer publish through one writer and one designer. Employees, agencies, freelancers and AI tools all produce posts, decks, emails, brochures and web pages at the same time. Each one interprets the brand guidelines a little differently, and the guidelines usually live in a PDF that nobody opens while writing. The damage shows up *after* publication: a wrong price, an outdated product claim, a banned term, an off-palette graphic, or a promise such as "best in India" that the company cannot prove.

The statement asks for a product that reviews content **before** it is published and checks it against approved guidelines, facts and assets. Reading it closely, we see five things it requires beyond a simple checker:

- **Grounded, not guessed.** Every finding must point to the guideline, fact or asset it comes from. An unsupported opinion from an AI model is exactly the kind of untrustworthy output the statement warns about.
- **Explain and suggest, not reject.** Each issue needs a reason and a fix that preserves what the creator was trying to say.
- **Humans stay in control.** Creators, managers and reviewers collaborate through comments, approvals and revision history. Suggestions never override decisions silently.
- **Learning over time.** Repeated mistakes and unclear guidelines should surface as insights, so the guidelines themselves improve.
- **Confidentiality and disclosure.** Brand material is sensitive, and the use of AI or any external service must be stated clearly to users.

The specific part we will solve

We will build the full loop for **text content and images, plus PDF/DOCX documents**: import the brand knowledge base, review a draft, show explained findings with source references and suggested fixes, run a role-based approval workflow with version history, and show a governance insights dashboard. Video and audio are out of scope for this event (see Section 8).

2. Users Affected by the Problem

Organisational setting: a marketing or communications team of 5-50 people inside a company, college body or agency, where several creators publish across channels and one or two managers are accountable for brand reputation.

User	Pain points today	What they need from BrandLens
Content creators (employees, interns, freelancers, agency staff)	Do not remember every rule; guidelines are long PDFs; feedback arrives late as vague comments such as "this feels off-brand".	Instant, specific feedback while drafting, with a reason and a suggested fix they can accept or change.
Brand / marketing managers	Reviewing everything manually is slow; the same mistakes recur; no record of who approved what.	A review queue, approvals with history, and insights on repeated issues.
Authorised reviewers (legal, compliance, product)	Asked to check facts and claims by email or chat, with no context and no trail.	Claims routed to them with the exact source and the AI's reasoning, and a clear approve/reject action.
Brand administrators	Guidelines are scattered, outdated or ambiguous and nobody knows which parts cause confusion.	One place to maintain guidelines, approved facts, terminology and assets, plus a list of unclear rules.
Customers and the public (indirect)	Meet wrong, inconsistent or exaggerated messaging.	Trustworthy, consistent communication.

3. Proposed Solution and What Makes It Different

BrandLens is a web application with role-based workspaces. An administrator loads the organisation's approved knowledge: brand guidelines, an approved facts sheet (prices, product names, dates, claims that have been verified), a terminology list (preferred and banned terms) and reusable assets (logo, colour palette, fonts). Creators submit a draft;

BrandLens extracts the content, retrieves the relevant approved material, runs checks and returns a list of findings. Each finding contains:

- **What** is wrong, highlighted in the content;
- **Why** it was flagged, with the **source clause** (for example "Tone guide, section 2.3" or "Approved facts: warranty = 2 years");
- **Severity** (blocking / should fix / suggestion) and a **confidence level**;
- A **suggested correction** that keeps the creator's meaning, shown as a before/after the creator can accept, edit or dispute.

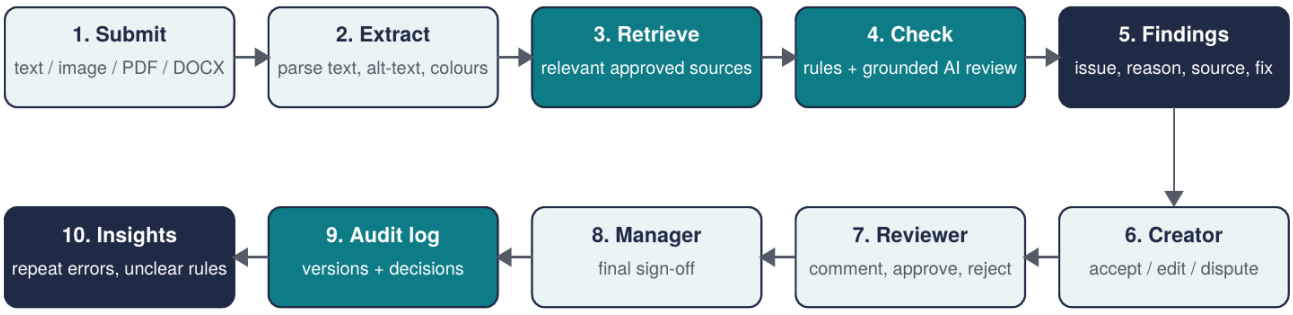
Value proposition

Brand problems are caught while they are cheap to fix, reviewers spend time only on real judgement calls, and the organisation gets a record of every decision plus data on where its guidelines fail.

What makes our approach different

Design choice	Why it matters
Source-first findings. A finding is shown only if it can cite a real clause, fact or asset from the organisation's knowledge base.	Prevents invented rules; makes every result checkable by a human in one click.
Deterministic checks for what can be computed (banned terms, price/date matching, colour palette distance, WCAG contrast ratio, missing alt text); AI only for judgement such as tone and claim support.	Reliable and repeatable results; AI cost and error are limited to the places where it is actually needed.
Dispute button. Creators can contest a finding with a reason; reviewers decide.	Keeps humans in control and turns disputes into data about unclear guidelines.
Guideline-gap detection. Findings that are frequently disputed or overridden mark a guideline as probably unclear.	Fixes the root cause, not just the symptoms. Most checkers stop at flagging content.
Intent-preserving rewrites with a diff view.	Creators keep ownership of the message; nothing is replaced automatically.

4. Complete Product Workflow



Human decides at every approval step. AI only suggests; nothing is auto-published or auto-rewritten. A rejected or edited draft returns to step 1 as a new version, with the full history kept.

Step	What happens
1. Set up knowledge base	Admin creates a workspace, uploads guidelines (PDF/DOCX/text), adds approved facts, terminology (preferred / banned terms) and assets (logo, palette). Documents are split into numbered clauses so findings can cite them.
2. Submit content	Creator pastes text or uploads an image, PDF or DOCX, chooses the channel (social post, email, web page, deck) and submits as Version 1.
3. Extract	Text is parsed; for images, dominant colours, size and presence of alt text are read; the content is split into reviewable pieces (sentences, claims, images).

Step	What happens
4. Retrieve and check	Relevant clauses and facts are retrieved for each piece. Deterministic rules run first, then the AI reviews tone, terminology, messaging and claim support using *only* the retrieved sources.
5. Validate findings	Each AI finding is checked: it must cite a retrieved source, return valid structured output and fit the schema. Findings that fail are dropped instead of shown.
6. Creator response	Creator sees highlighted issues with reasons, sources and suggested fixes, then accepts, edits or disputes each one and resubmits as a new version.
7. Review and approval	Content goes to a reviewer (comments, request changes, approve) and then to the manager for sign-off. Claims with no supporting fact are routed to an authorised reviewer.
8. Audit trail	Every version, finding, decision, comment and the AI services used are logged with who and when.
9. Insights	Dashboard shows the most common issue types, repeat offenders by team or channel (not blame, only patterns), and guidelines with frequent disputes or overrides.

5. Main Features We Intend to Build

Area	Features	Priority
Knowledge base	Import guidelines (PDF/DOCX/text); approved facts; terminology with preferred and banned terms; brand assets and palette; clause numbering and versioning.	Core (Round 2)
Content review	Text, image and PDF/DOCX review; checks for tone, terminology, factual accuracy, unsupported claims, colour palette use and accessibility (alt text, contrast).	Core (Round 2)
Explainable findings	Highlighted issue, reason, source clause, severity, confidence, suggested fix with before/after view.	Core (Round 2)
Human control	Accept / edit / dispute per finding; suggestions never auto-applied; manager can override any AI result with a reason.	Core (Round 2)
Workflow	Roles (admin, creator, reviewer, manager); comments; approvals; request-changes; revision history and diff.	Round 2-3
Insights	Recurring issue types, trend over time, most-disputed guidelines flagged as unclear.	Round 3
Trust and privacy	Per-organisation data isolation; visible AI-use disclosure; audit log; option to switch off external AI so only deterministic checks run.	Round 2-3

Three improvements we commit to demonstrate beyond the basic flow

1. **Guideline-gap detection** from disputed and overridden findings. 2. **Intent-preserving rewrite suggestions** with a diff view. 3. **Insights dashboard** for repeated mistakes over time.

6. Technology Stack, Architecture and Responsible Design

Layer	Technology	Why we chose it
Frontend	React, Vite, Tailwind CSS, Redux Toolkit	Our existing strength; fast to build the review screen with highlights, diffs and comments.
Backend	Node.js, Express.js, JWT auth with role-based access	One language across the stack; simple REST APIs; roles map directly to the workflow.
Database	MongoDB with Mongoose	Flexible documents suit content, findings and version history; holds clause embeddings for retrieval at hackathon scale.
Document parsing	pdf-parse, mammoth (DOCX), sharp (image data)	Open-source libraries that run on our own server, so documents are not sent elsewhere just to be read.

Layer	Technology	Why we chose it
Retrieval	Text embeddings stored per clause, cosine-similarity search	Lets the reviewer see only the relevant guidelines for each sentence, which limits hallucination.
AI model	Gemini API (free tier) for tone, claim and rewrite judgement, returning structured JSON	Accessible to students; JSON output can be validated before display. Model is behind one interface so it can be swapped.
Deterministic checks	Custom rule engine: term lists, regex for prices and dates, colour distance, WCAG contrast	Computed results are repeatable and free, and need no AI.
Deployment	Render (backend), Vercel (frontend), MongoDB Atlas	Free tiers, simple GitHub-based deploys.

The most technically challenging part, and how we address it

Making AI findings trustworthy and explainable. A language model asked "is this on brand?" will sound confident and sometimes be wrong or invent rules. We handle this with a *retrieve, review, verify* pipeline: (1) retrieve only the clauses and facts relevant to each piece of content; (2) require the model to answer in a fixed JSON schema that includes the clause ID it relied on and the exact text span in the draft; (3) verify in code that the clause ID exists in the retrieved set and the span exists in the draft, and discard any finding that fails; (4) run all computable checks outside the model; (5) show a confidence level and let a human dispute. We will build a small test set of good and bad sample content so we can measure how often findings are correct and report it honestly in Round 3.

Responsible design: privacy, security and human oversight

- **Confidentiality:** every record is scoped to an organisation ID; uploaded files are stored privately, never public; access is role-based; passwords hashed with bcrypt.
- **Disclosure:** a visible notice states when content is sent to an external AI service (Gemini), what is sent (only the relevant draft text and retrieved clauses) and that it is optional. Organisations can turn external AI off and still use the deterministic checks.
- **Data minimisation:** only the pieces needed for a check are sent to the model; confidential files are not shared in full.
- **Human oversight:** AI suggests, humans decide. Approvals can only be given by people with the right role, and every override and approval is logged.
- **Fairness and accessibility:** accessibility checks (alt text, contrast) are part of the product, and the interface itself will meet basic accessibility practice such as keyboard use and readable contrast.
- **Honest limits:** a clear note that BrandLens is an aid and not a legal compliance guarantee.

7. Responsibilities of Each Team Member

[Replace the Member 2 name with your teammate's actual name. The split below suits a 2-member team.]

Member	Role	Responsibilities
Divyanshu Kumar	Frontend and UI lead	React app structure and Redux store; knowledge-base screens; submission screen; review screen with highlighted issues, source panel and before/after diff; comments and approval UI; insights dashboard; accessibility of the interface; client-side API integration.
Anant Singh	Backend and AI pipeline lead	Express server, authentication and roles; MongoDB schemas; document parsing; clause splitting and embeddings; retrieval; deterministic rule engine; Gemini integration with JSON validation; workflow and audit-log APIs; deployment.
Both	Shared	Test set of sample brand guidelines and content; reviewing each other's code; end-to-end testing; documentation; user testing with 3-5 people; demo and presentation.

8. Development Plan Across Three Rounds (Feasibility and Scope)

Round	Goal	Deliverables
Round 1 (now)	Plan and validate	This document: problem understanding, users, solution, workflow, stack, risks and roles. Short interviews with 3-5 content publishers.
Round 2	Working core workflow end to end	Setup and deployment; login with roles; knowledge-base import; text review with source-cited findings; deterministic checks; findings screen with accept / edit / dispute; first version of the three named improvements started (rewrite with diff).
Round 3	Complete, reliable, polished	Image and PDF/DOCX review; full approval workflow with comments and revision history; insights dashboard and guideline-gap detection; AI-disclosure and external-AI off switch; accuracy test results; user-testing changes; documentation, limitations and roadmap; demo video.

Scope control. The first thing we make work is the shortest complete path: admin adds guidelines, creator submits text, findings appear with sources, reviewer approves. Images, documents and insights come only after this path works reliably. We deliberately leave out video and audio review, automatic publishing to social platforms, and custom model training. We reuse existing libraries and a hosted model instead of building our own, so two people can finish within the build window.

Risk	Mitigation
AI gives wrong or invented findings	Source-ID and text-span verification; deterministic checks first; confidence levels; dispute button; test set to measure accuracy.
Free-tier API limits or downtime	Deterministic checks keep working without AI; model behind one interface; results cached per version.
Time pressure with two people	Strict priority order above; integrate frontend and backend early and often.
No real brand data to test with	We create a sample brand (guidelines, facts, terms, assets) for a fictional or consenting organisation and disclose it.

9. Expected Impact

- **Creators** get clear feedback in seconds instead of days, and learn the guidelines as they work.
- **Managers and reviewers** spend time on real judgement calls, not on spotting a banned word or an old price.
- **Organisations** prevent public mistakes, keep an audit trail of approvals, and improve unclear guidelines with evidence.
- **Trust:** consistent and supportable claims protect credibility with customers.
- **Measurable in our demo:** issues caught before publication, share of findings accepted vs disputed, and repeat-issue rate across versions.

10. Future Scope After the Event

- Video and audio review (transcription plus frame checks); more file formats.
- Direct integrations with Google Drive, Slack, Notion and social scheduling tools for review before posting.
- Browser or document-editor plug-ins that review while the creator types.
- Organisation-specific tuning using past approved and rejected content; multilingual review.
- Self-hosted or on-premise model option for organisations that cannot send content to external services.
- Public-claim monitoring that rechecks already published content when a fact changes.

11. Complete Team and Participant Details

Name	Role in team	Institution / Program	Email	Phone
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